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Juhyun Lee, Ph.D.

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Research Interests

Soft implantable and wearable devices
Neural interfaces for sensing and stimulation
Wireless bioelectronic systems

Education

KAIST Daejeon, Korea
Ph.D. in Electrical Engineering (GPA: 4.08/4.3) Mar. 2019 – Aug. 2025
Thesis: Development of wireless optoelectronic brain implants and systems for broad applications in optogenetics

KAIST Daejeon, Korea
B.S. in Electrical Engineering (Cum Laude, GPA: 3.79/4.3) Mar. 2015 - Feb. 2019

Work Experience

Postdoctoral Researcher Daejeon, Korea
School of Electrical Engineering, KAIST Sep. 2025 – Present
Advisor: Prof. Jae-Woong Jeong

Visiting Student Researcher Stanford, CA, USA
Department of Chemical Engineering, Stanford University Sep. 2022 - Feb. 2023
Advisor: Prof. Zhenan Bao

Honors and Awards

BK21 Postdoctoral Fellowship, KAIST Sep. 2025 – Present
Government-Sponsored Scholarship, KAIST Mar. 2019 – Aug. 2025
Graduate Student Outstanding Paper Award, KAIST Nov. 2024
BK21 Financial Support for Overseas Long-Term Training, KAIST Sep. 2022 – Feb. 2023
Outstanding Teaching Assistant Award, KAIST Apr. 2021
National Scholarship for Science and Engineering, KOSAF Mar. 2015 - Dec. 2018

Publications

[†] These authors contributed equally to the work.

[1] C. Y. Kim†, **J. Lee†**, E. Y. Jeong, Y. Jang, H. Kim, B. Choi, D. Han, Y. Oh, J.-W. Jeong (2025). "Wireless technologies for wearable electronics: A review". ***Advanced Electronic Materials**, 11(10), 2400884* **[Featured on the Front Cover]**

[2] H. Kim†, **J. Lee†**, U. Heo, D. Jayashankar, K.-C. Agno, Y. Kim, C. Y. Kim, Y. Oh, S.-H. Byun, B. Choi, H. Jeong, W.-H. Yeo, Z. Li, S. Park, J. Xiao, J. Kim, J.-W. Jeong (2024). "Skin preparation-free, stretchable microneedle adhesive patches for reliable electrophysiological sensing and exoskeleton robot control". ***Science Advances**, 10(3), eadk5260*

[3] K. E. Parker†, **J. Lee†**, J. R. Kim†, C. Kawakami, C. Y. Kim, R. Qazi, K.-I. Jang, J.-W. Jeong, J. G. McCall (2023). "Customizable, wireless, and implantable optogenetic probe design and fabrication via 3D printing". ***Nature Protocols**, 18(1), 3-21* **[Featured on the Cover]**

[4] **J. Lee†**, K. E. Parker†, C. Kawakami†, J. R. Kim†, R. Qazi, J. Yea, S. Zhang, C. Y. Kim, J. Bilbily, J. Xiao, K.-I. Jang, J. G. McCall, J.-W. Jeong (2020). "Rapidly customizable, scalable 3D-printed wireless optogenetic probes for versatile applications in neuroscience". ***Advanced Functional Materials**, 30(46), 2004285* **[Featured on the Back Cover)**

[5] C. Y. Kim†, K. E. Parker†, E. Y. Jeong, D. Han, Y. Jang, S.-Y. Joo, **J. Lee**, C. Chiangchaisakunthai, R. Qazi, J. Chung, S. Ha, J.-W. Jeong, J. G. McCall (2026). "Protocol to design and implement scalable wireless network system for high-throughput, automated behavioral and circuit neuroscience in mice". ***STAR Protocols**, 7(2), 104453*

[6] I. Kang†, J. Bilbily†, C. Y. Kim, C. Shi, M. K. Madasu, E. Y. Jeong, K. E. Parker, D. A. Kwon, B.-J. Jung, J.-S. Yang, **J. Lee**, N. D. L. Kabbaj, W. Lee, J.-B. Yoon, R. Al-Hasani, J. Xiao, J. G. McCall, J.-W. Jeong (2025). "Wireless modular implantable neural device with one-touch magnetic assembly for versatile neuromodulation". ***Advanced Science**, 12(4), 2406576*

[7] S. Oh†, S. Lee†, S. W. Kim, C. Y. Kim, E. Y. Jeong, **J. Lee**, D. A. Kwon, J.-W. Jeong (2024). "Softening implantable bioelectronics: material designs, applications, and future directions". ***Biosensors and Bioelectronics**, 116328*

[8] G.-H. Lee†, H. Kim†, **J. Lee**, J.-Y. Bae, M. Kim, C. Yang, S. Park, H. Kang, S.-K. Kang, J. Kang, Z. Bao, J.-W. Jeong, S. Park (2023). "Large-area photo-patterning of initially conductive EGaIn particle-assembled film for soft electronics". ***Materials Today**, 67, 84-94*

[9] S.-H. Byun†, J. Y. Sim†, Z. Zhou, **J. Lee**, R. Qazi, M. C. Walicki, K. E. Parker, M. P. Haney, S. H. Choi, A. Shon, G. B. Gereau, J. Bilbily, S. Li, Y. Liu, W.-H. Yeo, J. G. McCall, J. Xiao, J.-W. Jeong (2019). "Mechanically transformative electronics, sensors, and implantable devices". ***Science advances**, 5(11), eaay0418*

Conference Presentations

[1] **J. Lee**, K. E. Parker, J. G. McCall, J.-W. Jeong (2026). "Wireless, spatially-matched optoelectronic neural interfaces for simultaneous optogenetics and electrophysiology". ***The 28th Korean MEMS Conference**, invited*

[2] **<u>J. Lee</u>**, K. E. Parker, J. G. McCall, J.-W. Jeong (2025). "Spatially-matched optoelectronic neural probes for simultaneous optogenetics and electrophysiology". ***2025 Fall Conference of the Korean Sensors Society***, *poster presentation*

[3] **<u>J. Lee</u>**, K. E. Parker, J. G. McCall, J.-W. Jeong (2024). "Wireless optoelectronic neural interfaces for simultaneous and spatially-matching optogenetics and electrophysiology". ***The 12th World Biomaterials Congress***, *poster presentation*

[4] **<u>J. Lee</u>**, K. E. Parker, C. Kawakami, J. R. Kim, R. Qazi, J. Yea, S. Zhang, C. Y. Kim, J. Bilbily, J. Xiao, K.-I. Jang, J. G. McCall, J.-W. Jeong (2021). "Customizable and rapidly manufacturable 3D-printed neural probes for wireless optogenetics". ***2021 Virtual MRS Spring Meeting & Exhibit***, *oral presentation*

[5] H. Kim, **<u>J. Lee</u>**, U. Heo, D. Jayashankar, K.-C. Agno, Y. Kim, C. Y. Kim, Y. Oh, S.-H. Byun, B. Choi, H. Jeong, W.-H. Yeo, Z. Li, S. Park, J. Xiao, J. Kim, J.-W. Jeong (2024). "Skin-preparation-free, stretchable microneedle adhesive patches for high-fidelity electrophysiological sensing and exoskeleton robot control". ***2024 MRS Spring Meeting & Exhibit***, *oral presentation*

[6] H. Kim, **<u>J. Lee</u>**, J.-W. Jeong (2022). "Adhesive, air-permeable, stretchable conductive sensors for long-term electrophysiological signal monitoring". ***The 6th International Conference on Active Materials and Soft Mechatronics***, *poster presentation*

[7] S.-H. Byun, J. Y. Sim, Z. Zhou, **<u>J. Lee</u>**, R. Qazi, M. C. Walicki, K. E. Parker, M. P. Haney, S. H. Choi, A. Shon, G. B. Gereau, J. Bilbily, S. Li, Y. Liu, W.-H. Yeo, J. G. McCall, J. Xiao, J.-W. Jeong (2020). "Mechanically transformative electronics enabled by Gallium-elastomer composite". ***2020 BMES Annual Meeting***

Teaching Experience

Research Mentorship, KAIST

Mentored 6 undergraduate and 3 graduate students — Jun. 2019 - present

Teaching Assistant, KAIST

EE405 Electronics Design Lab (Device)	Spring 2020, Fall 2021, Spring 2022, Fall 2023
EE566 MEMS in EE Perspective	Spring 2021
EE405 Electronics Design Lab (Circuit)	Fall 2019

Freshman Tutoring Program, KAIST

General Physics I — Spring 2016, Fall 2017, Spring 2018

Extracurricular Experience

Futsal Club, KAIST	Sep. 2017 - Feb. 2019
Classical Guitar Club, KAIST	Mar. 2015 - Feb. 2019
Language Exchange Program (German), KAIST	Sep. 2017 – Dec. 2017
Exchange student, Karlsruhe Institute of Technology	Mar. 2017 - Aug. 2017